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Estimating age- and sex-specific mortality rates for small areas with TOPALS regression: An application to Brazil in 2010

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Background

- ❖ Estimates of levels, age patterns, and sex differences in small-area mortality are important for evaluating and targeting public investment
- ❖ Main problems for small-area mortality estimation:
 1. Small risk populations
 2. High sampling variability in recorded vital events
 3. Data quality (Completeness and Age Misreporting)

Background

- ❖ Classic indirect methods and their modern variants:
 - **Child Mortality:** (BRASS and COALE 1968; SULLIVAN 1972; FEENEY 1980; UNITED NATIONS 1983; HILL 1991; MOULTRIE et al. 2013)
- ❖ Countries with good vital registration systems and annual population updates:
 - **Modern Statistical Modeling:** (TOSON; BAKER, 2003; EAYRES; WILLIAMS, 2004; BRAVO; MALTA 2010; MCKINNON, 2010; TSIMBOS et al., 2014; ALEXANDER et al., 2016)

Background

- ❖ For developing countries (Brazil):
 - New methods for estimating small-area mortality and life expectancy: (LIMA, E.E.C.; QUEIROZ, B.L.; MISOV, T.; LENART, A., 2016)
- ❖ Common approach in Brazil:
 - Indirect Standardization: (FREIRE et al., 2015; LIMA, QUEIROZ & SAWYER, 2014; LIMA & QUEIROZ, 2014; UNDP, 2013; HORTA et al., 1998)
 - It requires a strong mathematical assumption about local age patterns

Main objective

- ❖ Poisson regression for log mortality at ages 0...99, for areas with very small populations
- ❖ TOPALS (De Beer, 2012):
 - Relational model
 - Age schedule = Standard Schedule + Age-Specific Offsets

Our version of TOPALS:

- ❖ We assume that the schedule of log mortality rates over ages (0,...,99) is the sum of a standard schedule λ^* and a linear spline function:

$$\underset{100 \times 1}{\lambda(\alpha)} = \underset{100 \times 1}{\lambda^*} + \underset{100 \times 7}{\mathbf{B}} \underset{7 \times 1}{\alpha}$$

- α : a vector of parameters representing offsets to the standard schedule $(\lambda_x - \lambda_x^*)$
- $\mathbf{B}$: a matrix of constants in which each column is a linear B spline basis function

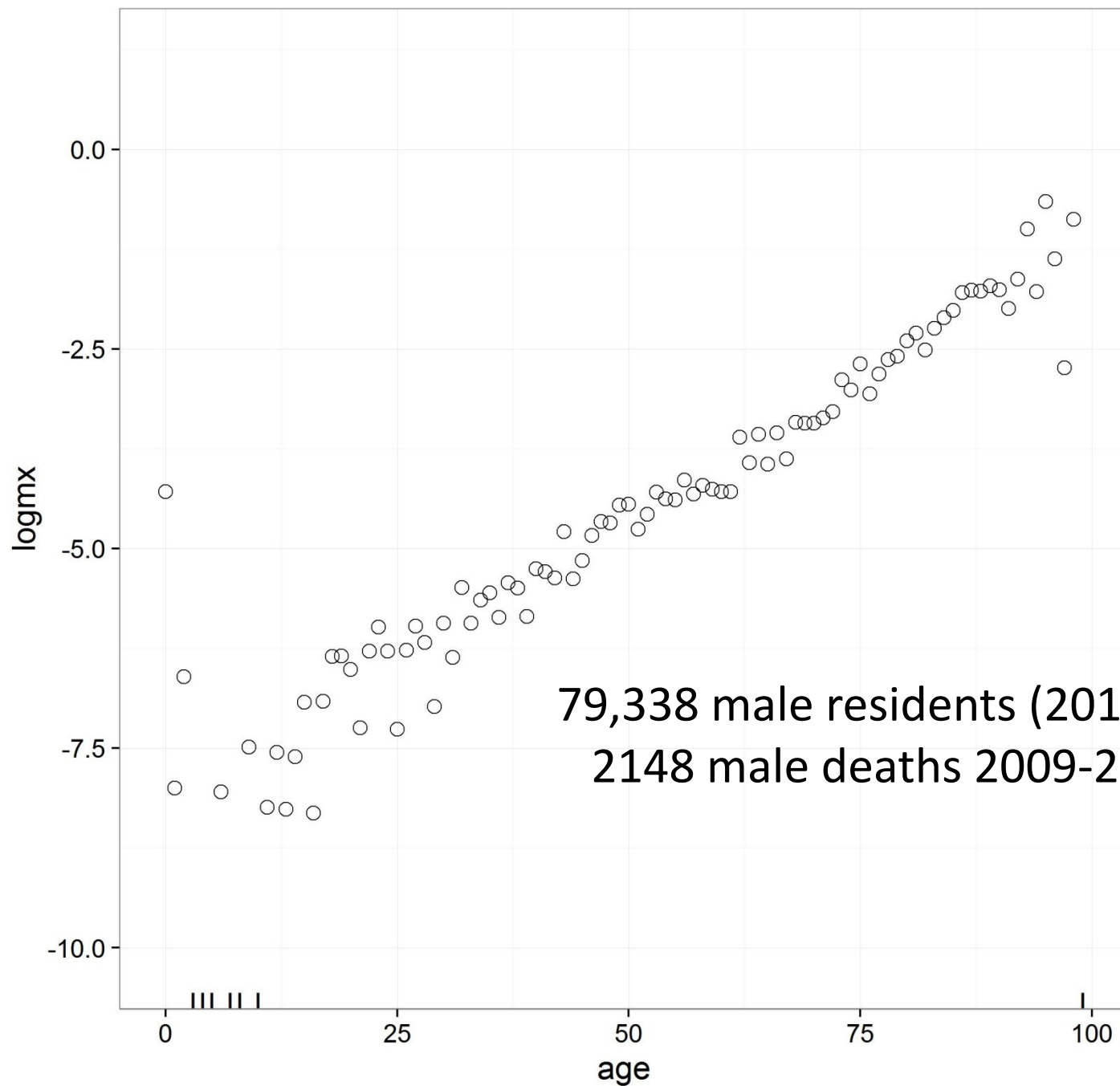
Data sources

- ❖ **Population:** Brazilian Demographic Census (2010)
- ❖ **Death:** Mortality Information System of the Ministry of Health (SIM/Datasus)
- ❖ **Standards:** Human Mortality Database (2015)

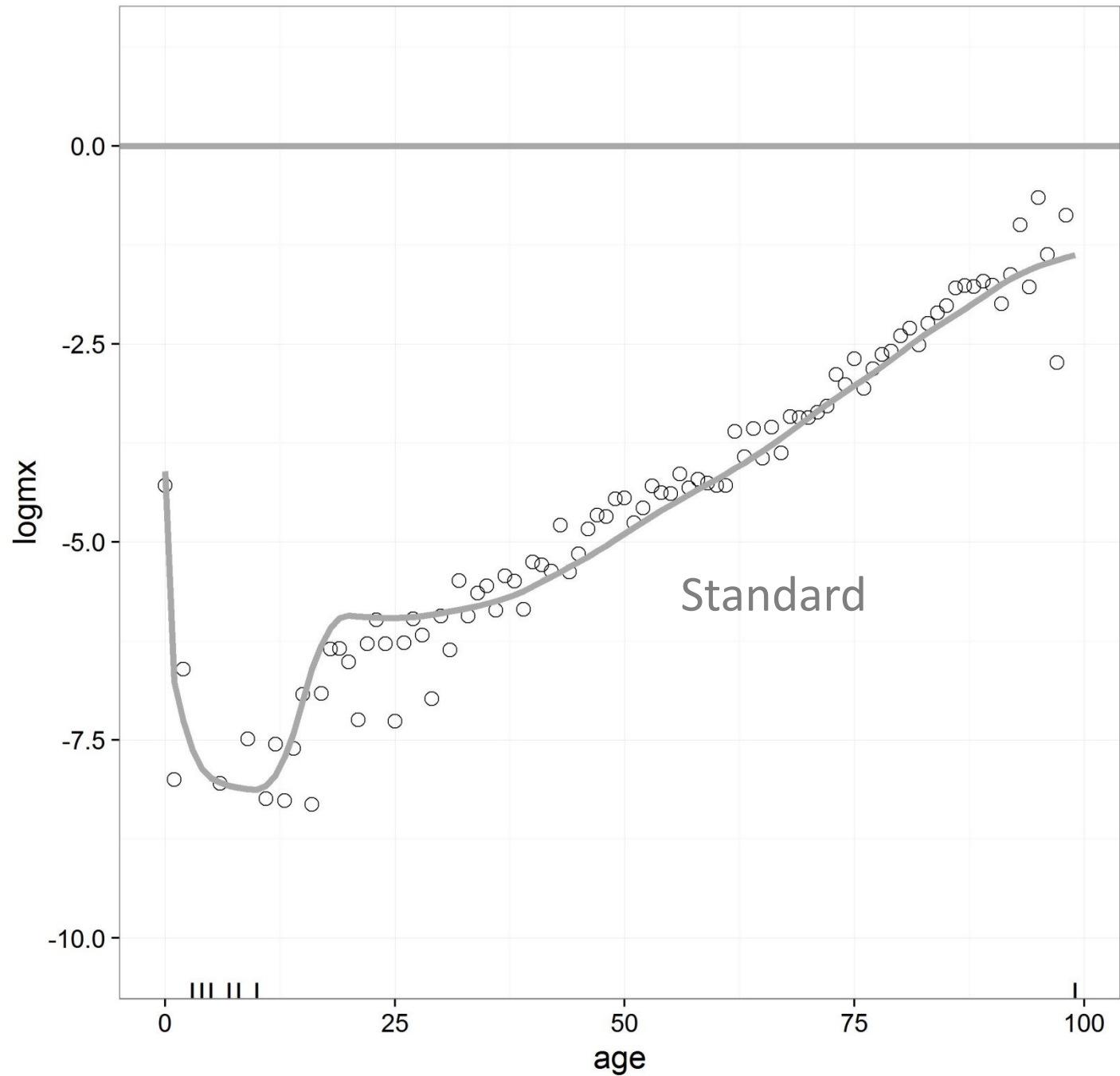
Database

State	Meso	Micro	Muni	Sex	Age	Pop 2010	Expos 2009:2011	Deaths 2009:2011
SP	Bauru	Lins	Lins	F	0	401	1176	21
:	:	:	:	:	:	:	:	:
SP	Bauru	Lins	Lins	F	99	3	10	5
SP	Bauru	Lins	Lins	M	0	432	1307	18
:	:	:	:	:	:	:	:	:
SP	Bauru	Lins	Lins	M	99	1	3	0
SP	Bauru	Lins	Guaimbê	F	0	41	115	0
:	:	:	:	:	:	:	:	:
1,113,000 combinations of (5.565 muni x 2 sexes x 100 ages)								

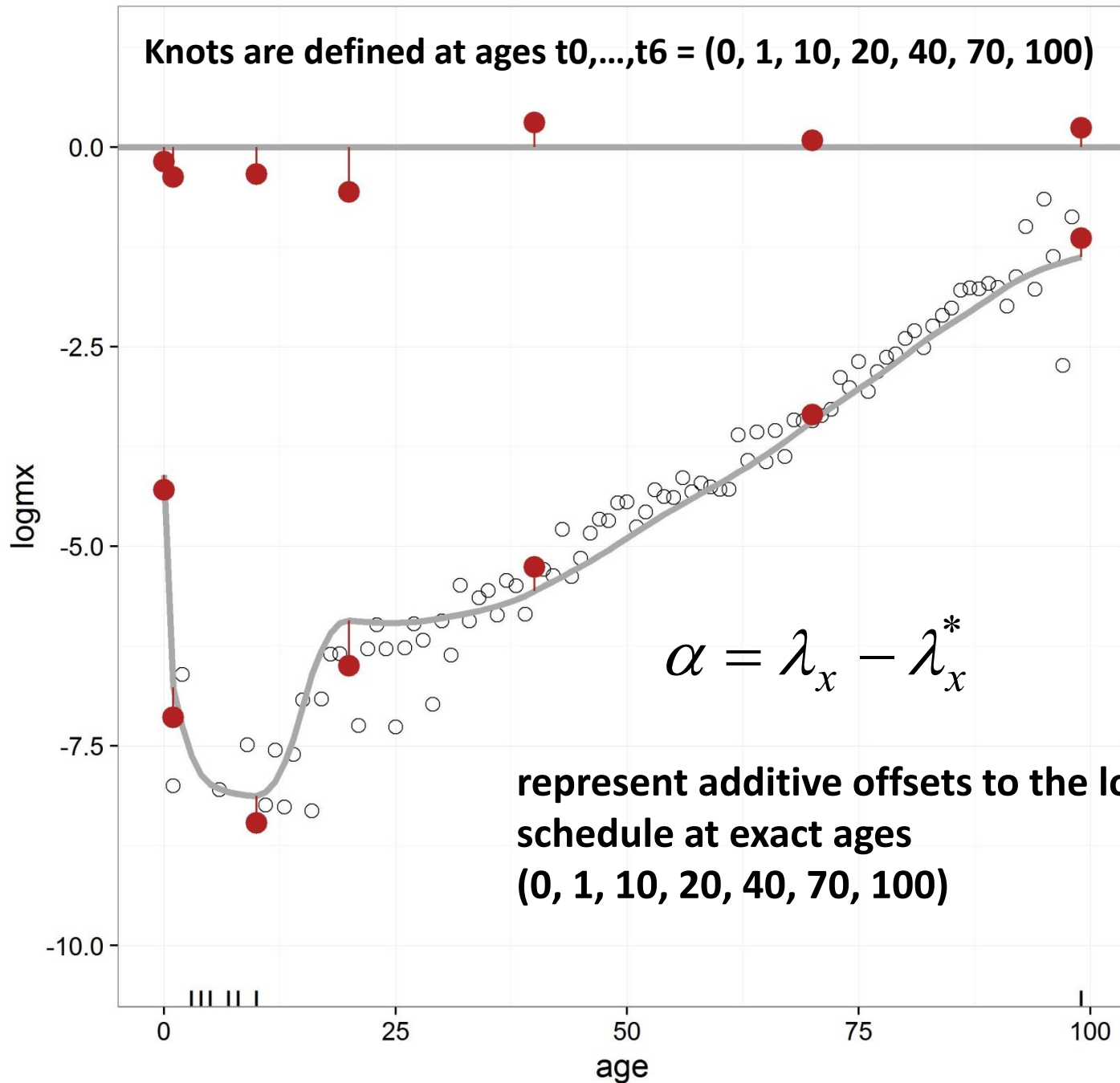
35019 Lins SP Males



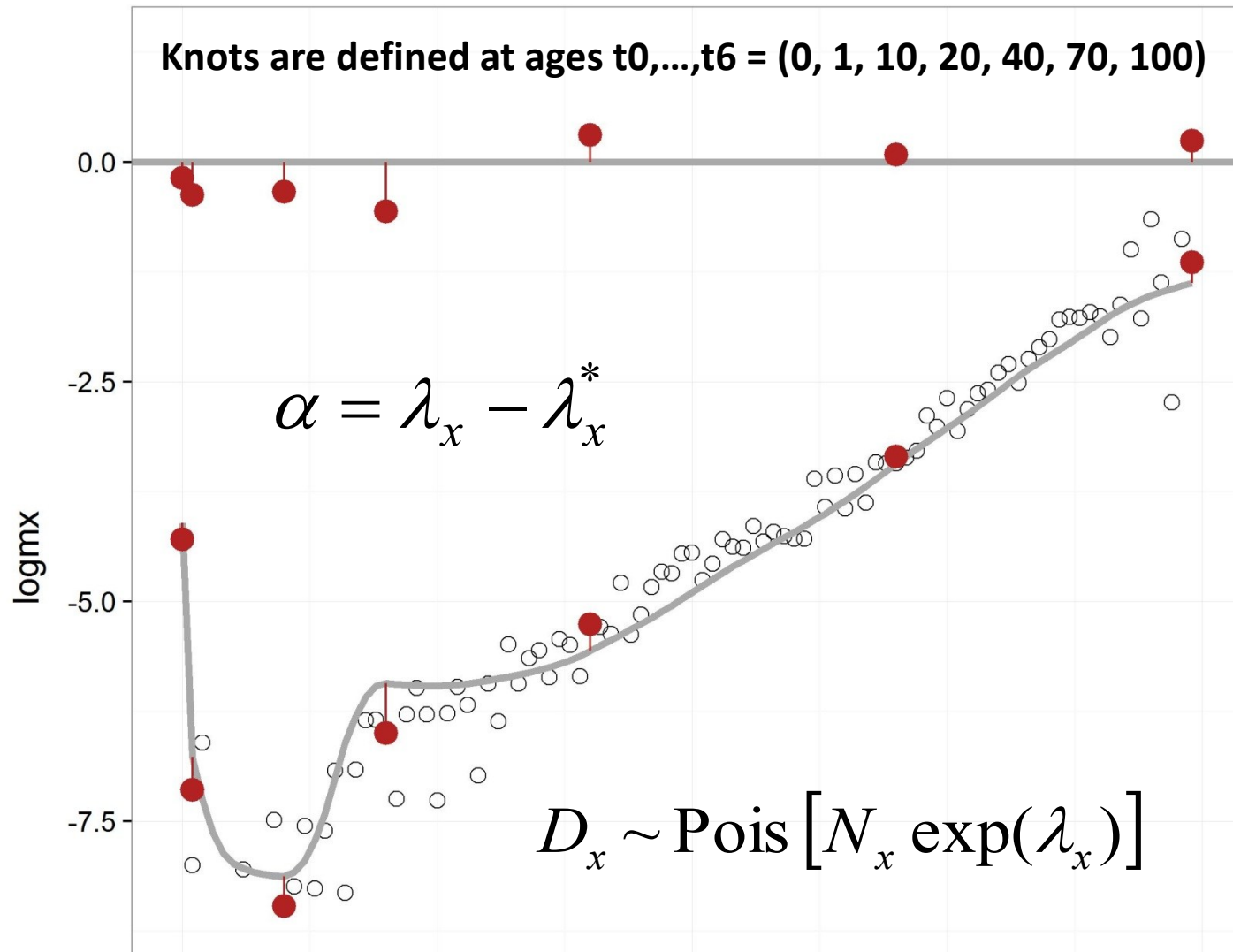
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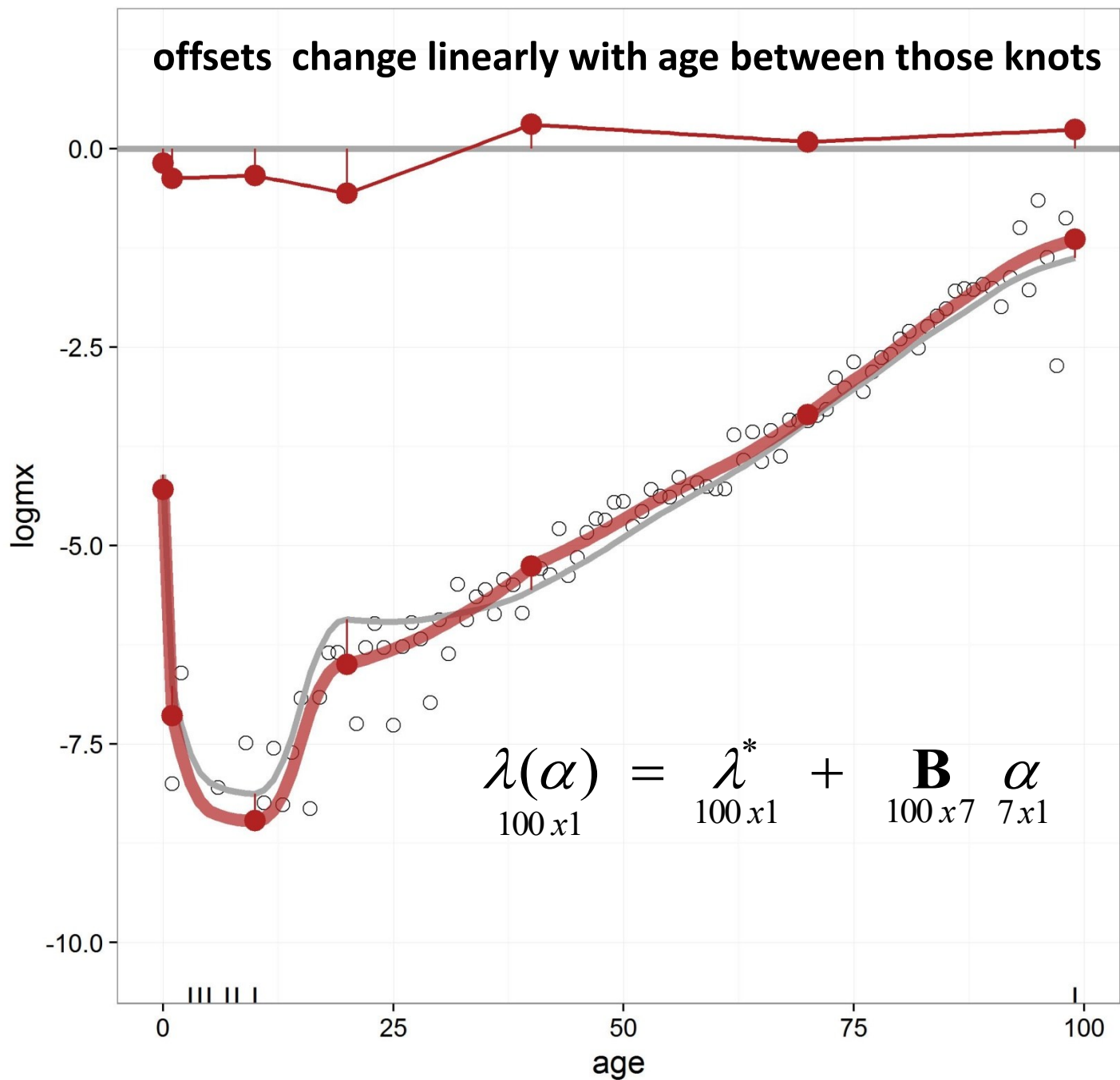


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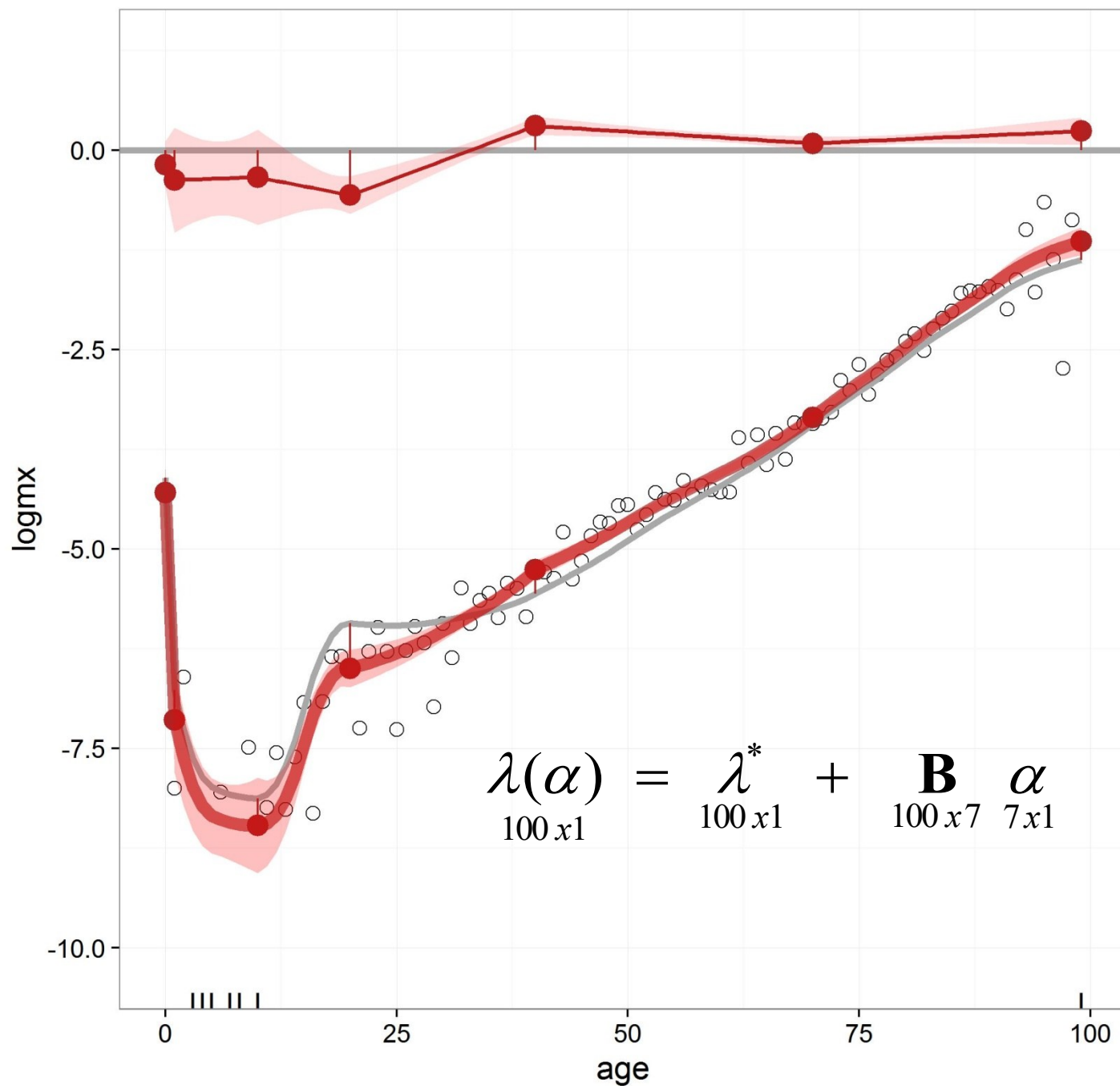


$$\log L(\alpha) = \text{constant} + \sum_x [D_x \lambda_x(\alpha) - N_x \exp(\lambda_x(\alpha))]$$

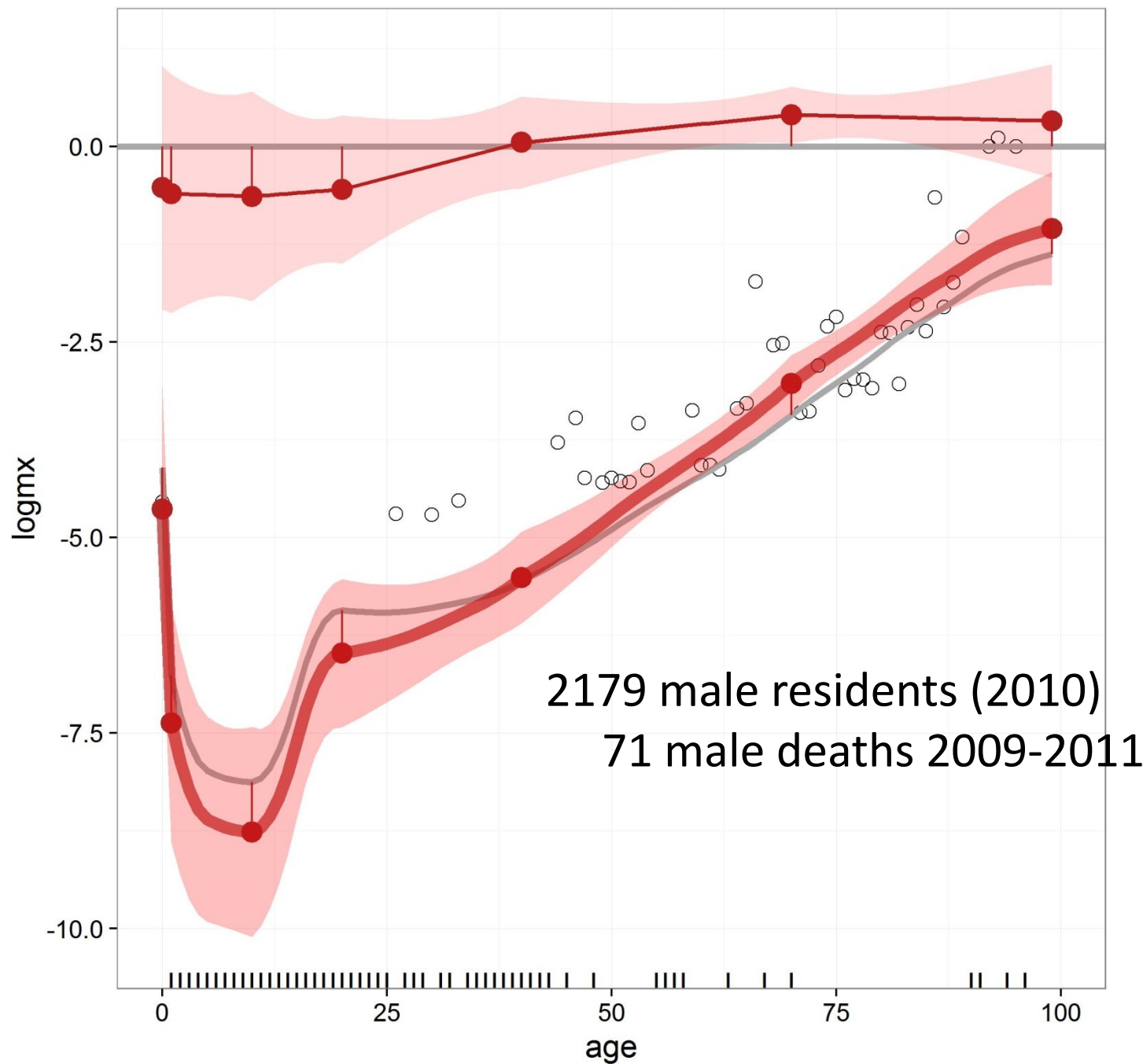
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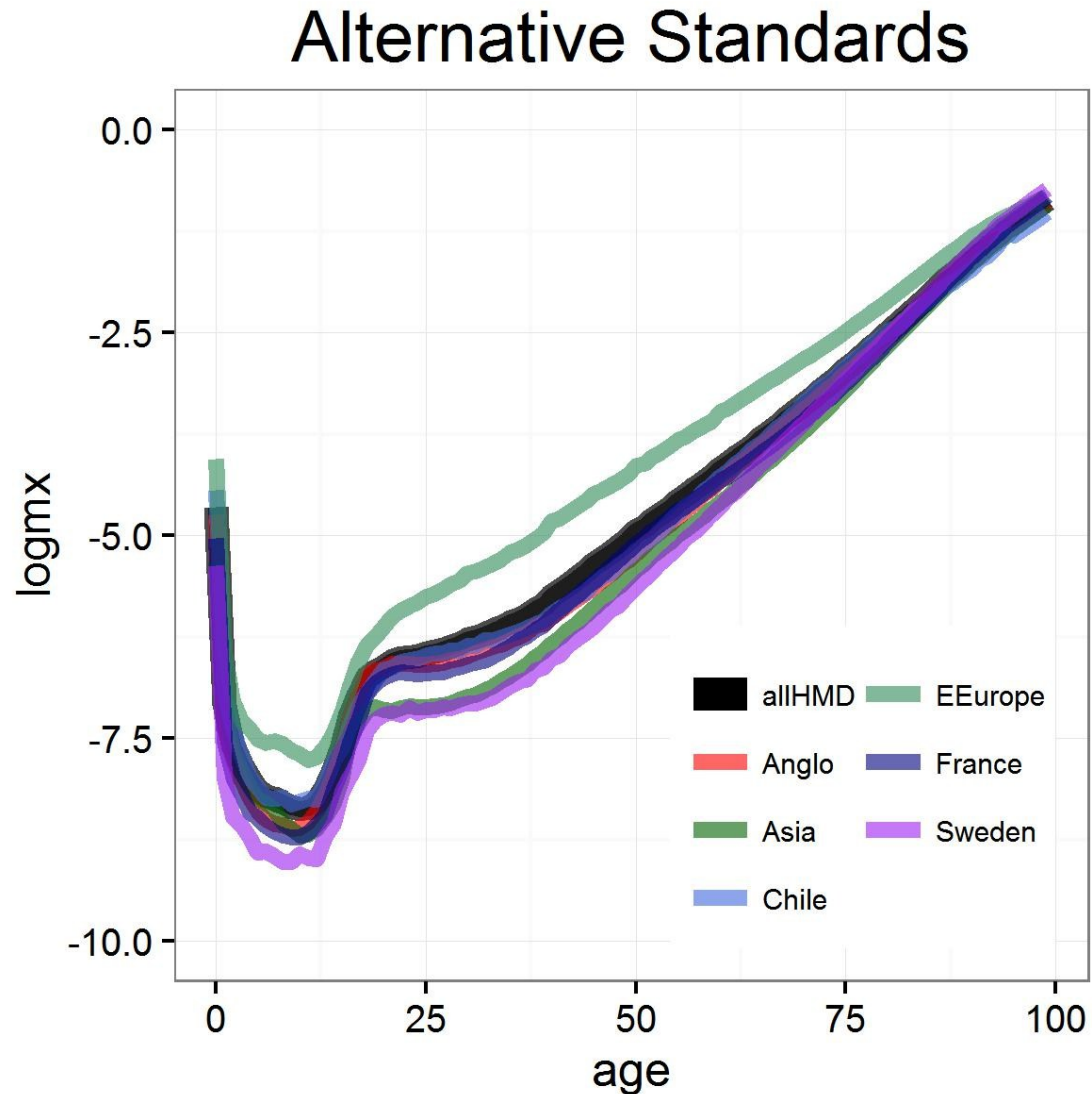
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352580 Júlio Mesquita Males



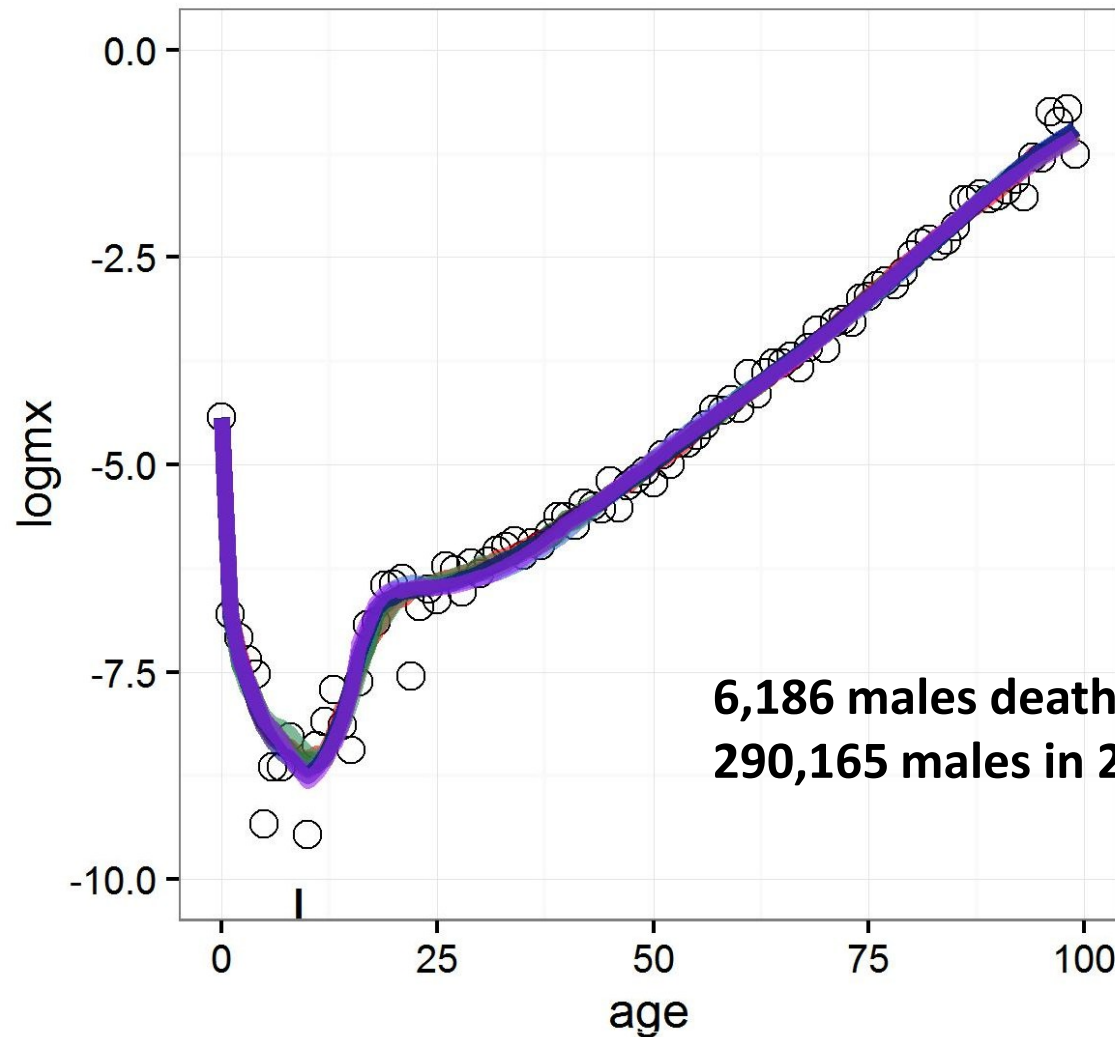
Insensitivity to choice of standard



Source: Human Mortality Database (2015)

Insensitivity to choice of standard

Ribeirão Preto

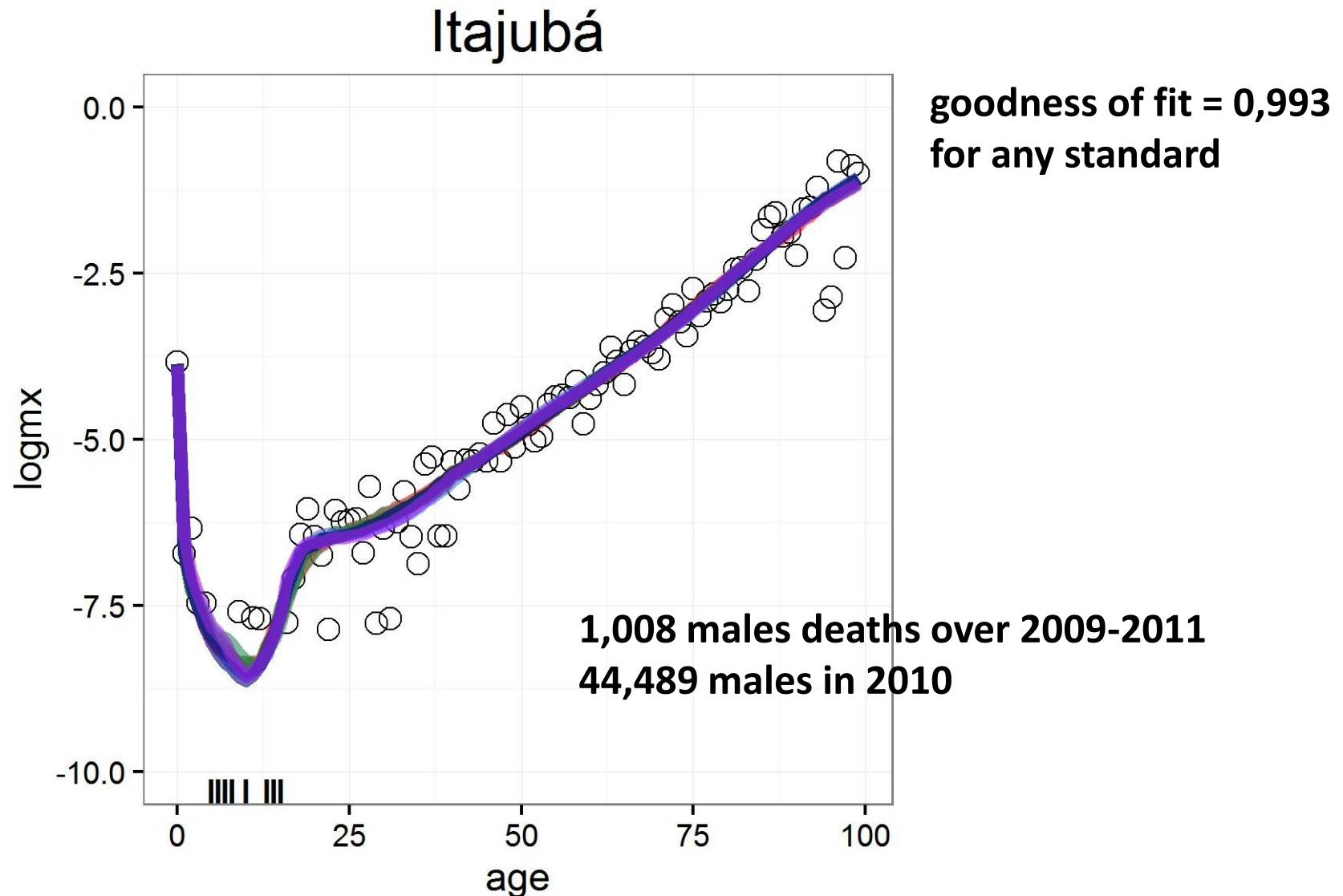


**goodness of fit = 0,999
for any standard**

**6,186 males deaths over 2009-2011
290,165 males in 2010**

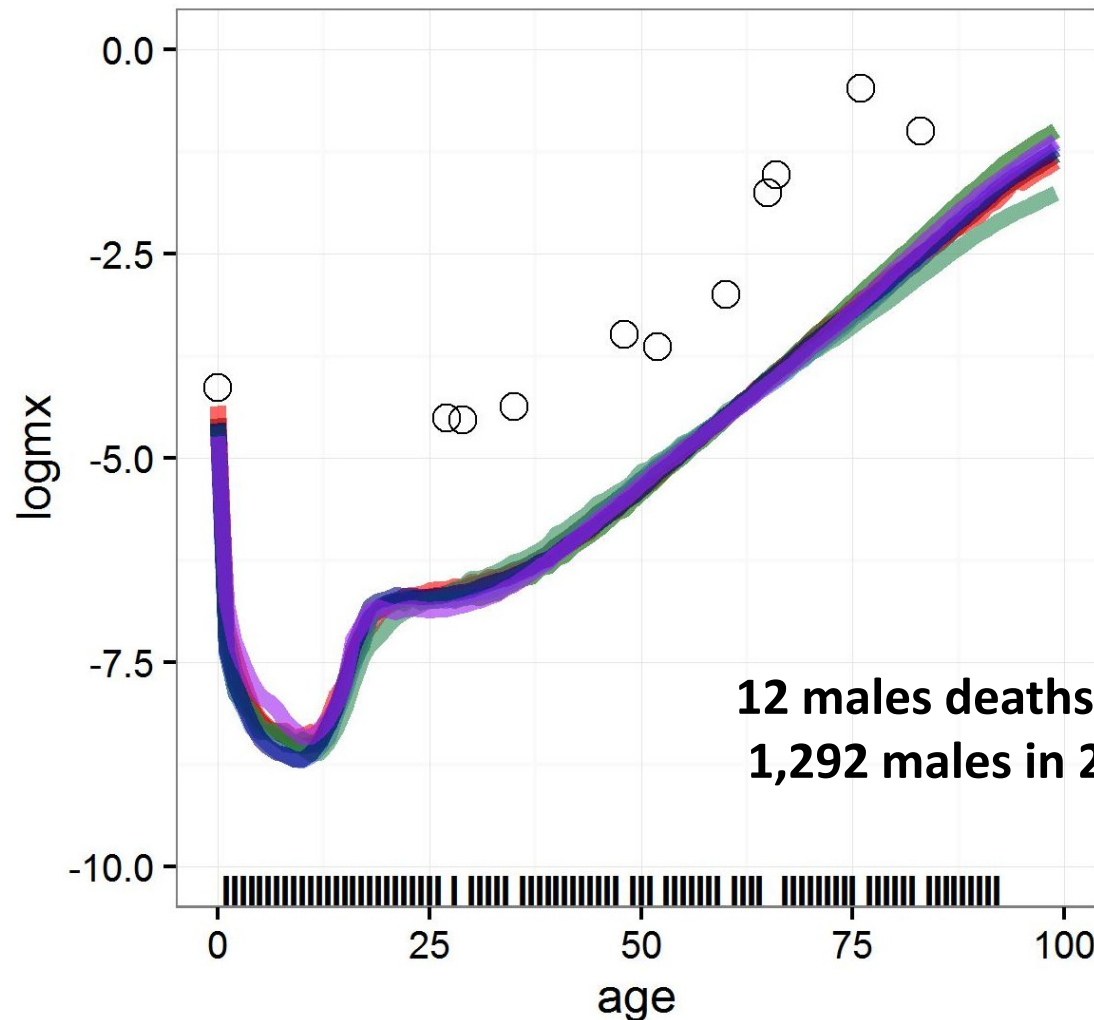
Source: IBGE (2010) and Ministry of Health

Insensitivity to choice of standard



Insensitivity to choice of standard

Fernando de Noronha



**goodness of fit =
from 0,670 to 0,689**

**12 males deaths over 2009-2011
1,292 males in 2010**

Source: IBGE (2010) and Ministry of Health

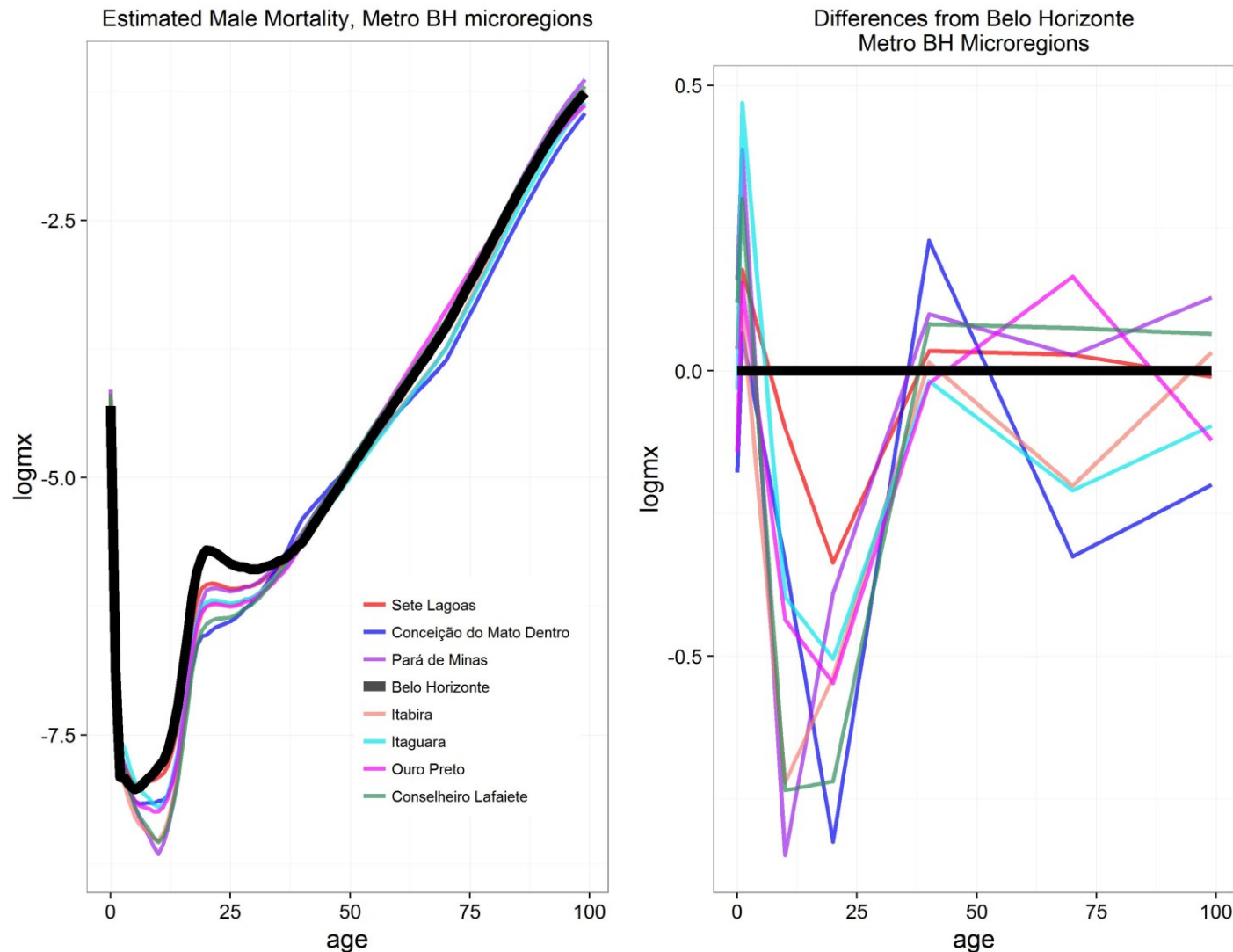
Estimation Errors for TOPALS and Indirect Standardization with Simulated Data

- ❖ Mean Absolute Error: Estimated Life Expectancy (e_0) over 5000 Monte Carlo Samples

N_x	<i>Indirect Standardization</i>		<i>TOPALS Regression</i>	
	<i>Correct Std</i>	<i>Incorrect Std</i>	<i>Correct Std</i>	<i>Incorrect Std</i>
1	4.82	6.34	4.98	6.12
100	0.52	5.54	1.27	1.44
10000	0.05	5.54	0.15	0.15

The underlying mortality pattern in Brazilian small areas

Male mortality in microregions of the RMBH-MG, 2010



Source: IBGE (2010) and Ministry of Health

Discussion/Conclusion

- ❖ The method offers at least three advantages over prior standard approaches to estimating small-area mortality schedules:
 - It uses data for single years of age 0,1,2,...,99, even when the corresponding risk populations are very small
 - It allows appropriate use of the “zero” cells in which there are no recorded deaths
 - Most importantly, it does not assume a specific age pattern of mortality rates in advance

Discussion/Conclusion

- ❖ The method offers an incomplete solution in areas with incomplete deaths records
- ❖ However, it is a useful method to estimate and smooth rates before applying DDM methods to estimate the undercount of deaths
- ❖ Present stage: integrate TOPALS regression with models of age-specific reporting errors
- ❖ Supplementary material will shortly be available on our project website

Obrigado!

<https://topals-mortality.schmert.net/>

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